Notes: Boone & Mulherin (JF, 2007)

How Are Firms Sold?

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1 The core question and why the paper matters

1.1 The measurement problem

A large empirical takeover literature proxies "auction" intensity by counting *public bids*. But public bids occur *late* in the takeover process and may dramatically understate competition if:

- the target (and its investment bank) contacts many potential buyers privately,
- many potential buyers sign confidentiality/standstill agreements and receive data rooms,
- a small number submit *private written offers*,
- only the eventual winner (and occasionally one rival) proceeds to a *public* bid stage.

1.2 The paper's key contribution

Boone and Mulherin (2007) open the "black box" of the **private takeover process** using the background sections of merger and tender-offer filings (e.g., 14A, S-4, 14D). They build new measures of competition:

$$\text{Contact}, \quad \text{Confidentiality}, \quad \text{Private written bids}, \quad \text{Public bids}.$$

This produces a different classification of *auctions vs. negotiations* than classifications based only on public bidder counts.

2 Institutional structure: the takeover timeline (Figure 1)

2.1 Two phases: private vs. public

The paper stresses that a takeover has a **private phase** and a **public phase**.

- **Private takeover process:** from the initiation of sale discussions to the signing of an agreement with a bidder.
- **Public takeover process:** from signing/announcement to completion (including any public contest).

2.2 Figure 1 (interpretation)

Figure 1 provides the timeline:

1. *Initiation event* (e.g., target hires an investment bank; CEOs meet; board explores "strategic alternatives").
2. *Private date:* start of the private process.
3. *Contact* and *confidentiality/standstill* agreements (data room access).
4. *Private written offers* arrive (often after one or more rounds).

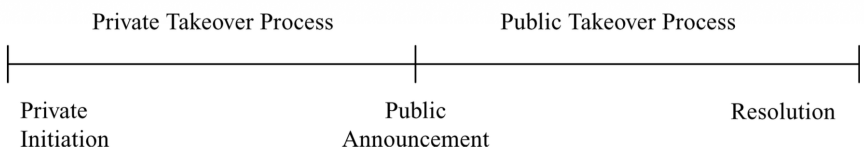

Figure 1. Timeline of the takeover process. This figure provides a timeline of the takeover process. The private takeover process is the period from the private initiation of the takeover to the first public announcement of the takeover. The public takeover process is the period from the first public announcement of the takeover to the resolution of the takeover.

5. *Rumor date* may occur before a formal agreement.
6. *Agreement date:* definitive merger agreement signed.
7. *Merger announcement:* public disclosure of the deal.
8. *Completion date:* closing (or deal withdrawal).

Why it matters empirically: if we measure "competition" only by *public* bids, we miss the earlier and often richer competitive process that determines which bidder becomes the "winner" that goes public.

3 Data and classification

3.1 Sample formation

- Sample: 400 takeover attempts of major U.S. corporations announced over 1989–1999.
- Composition: 377 completed and 23 withdrawn deals (high completion rate consistent with large targets).
- Sources: SEC EDGAR filings (1993+), and LexisNexis / Laser Disclosure (pre-1993).
- Sale-process information is manually extracted from the *background* sections of merger and tender-offer filings.

3.2 Process variables (key measurement objects)

For each target i, the paper records:

- Contact_i: number of potential buyers contacted by the target and its investment bank,
- Confidential_i: number of potential buyers signing confidentiality/standstill agreements,
- PrivateBid_i: number of potential buyers making private written offers,
- PublicBid_i: number of potential buyers making public offers reported in the financial media.

3.3 Auction vs. negotiation classification

$$\text{Auction}_i \equiv \mathbf{1}\{\text{multiple potential buyers are contacted and (typically) multiple sign confidentiality}\}. \tag{1}$$

Negotiations are processes focused on a single buyer (sometimes with minimal preliminary contact elsewhere).

Important nuance. The paper further distinguishes *formal* vs. *informal* auctions (structured multi-round vs. less structured), but shows that using a single auction category does not change the core findings.

4 Descriptive evidence: how firms are sold (Tables I–IV)

4.1 Table I: examples of the sales process

Table I provides two concrete illustrations:

- **Auction example:** the seller hires an investment bank and contacts many potential buyers (dozens), many sign confidentiality, a small number submit private bids, and only one may become a public bidder.
- **Negotiation example:** the process is centered on a single buyer from the outset (contact/confidentiality/private bid all near 1).

Interpretation: "one public bidder" does *not* imply "no competition"; competition often happens privately.

Table I
Examples of the Sales Process

This table provides examples of the auction and negotiation procedures by which firms are sold. *Auction* refers to cases in which the selling firm contacts multiple potential buyers. *Negotiation* refers to a sales process focusing on a single buyer. *Private Date* is the date on which the sales process begins. *Rumor Date* (if any) is the date on which the possibility of a takeover is mentioned in the financial media prior to a formal agreement date. *Agreement Date* is the date on which the seller and bidder sign the takeover agreement. *Merger Announce* is the date on which the financial media report the takeover agreement. *Completion Date* is the date on which the takeover is completed. *Contact* refers to the number of potential bidders contacted by the seller and its investment bank. *Confidentiality* refers to the potential buyers that engage in a confidentiality/standstill agreement. *Private Bidders* refers to the potential buyers that submit a private written offer. *Public Bidders* refers to the potential buyers that announce a formal bid for the seller in the financial media.

	Auction	Negotiation
Seller	Blount Inc.	BankBoston
Bidder	Lehman Bros	Fleet Financial
Initiation event	Target contacts investment bank	CEOs meet
Private date	8/26/1998	4/1/1998
Contact	65	1
Confidentiality	28	1
Private bidders	2	1
Public bidders	1	1
Rumor date	3/31/1999	n/a
Agreement date	4/18/1999	3/14/1999
Merger announce	4/20/1999	3/15/1999
Completion date	8/19/1999	10/1/1999

Table II
Summary of the Sales Process

This table summarizes the sales process for the sample of 400 takeovers. Panel A describes the sales process for the full sample. *Contact* reports the average number of potential buyers contacted by the selling firm and its investment bank. *Confidential* reports the average number of potential buyers that engage in a confidentiality/standstill agreement. *Private Bidders* reports the average number of potential buyers that submit a private written offer. *Public Bidders* reports the average number of potential buyers that announce a formal bid for the firm in the financial media. In Panel B, the aspects of the sales process are reported for auctions and negotiations. *Auction* refers to cases in which the selling firm contacts multiple potential buyers. *Negotiation* refers to a sales process focusing on a single buyer.

Sample	Observations	Contact	Confidential	Private Bidders	Public Bidders
Panel A: The Full Sample					
Full Sample	400	9.49	3.75	1.29	1.13
Panel B: The Sample Categorized by Sales Process					
Auction	202	20.67	6.83	1.57	1.24
Negotiation	198	1.02	1	1	1.02

4.2 Table II: summary of the sales process (private vs. public depth)

Table II aggregates the process measures.

Panel A (full sample). On average, targets contact about 9.5 potential buyers and obtain roughly 1.3 private written bids, yet the average number of *public* bidders is only about 1.13. **Interpretation:** public bid counts dramatically understate the depth of the private process.

Panel B (by sales procedure).

- In **auctions**, targets contact about 21 potential buyers and have about 6.8 confidentiality agreements.
- In **negotiations**, contact/confidentiality/private bids are essentially 1 by construction.

Interpretation: the paper's classification corresponds to a qualitatively different information-production process in auctions.

4.3 Table III: sample by year

Table III shows takeovers cluster in the second half of the 1990s, but auctions and negotiations have similar time patterns. **Interpretation:** results are not driven by a narrow time slice; both procedures are used throughout the sample.

Table III
The Sample by Year

This table reports the number of takeovers per year for the sample period of 1989 to 1999. Observations are placed in the year of announcement. Data are reported for the full sample and for the two sales procedures, auction and negotiation.

Year	Full Sample	Auction	Negotiation
1989	12	10	2
1990	34	16	18
1991	19	8	11
1992	21	11	10
1993	15	5	10
1994	30	16	14
1995	59	29	30
1996	39	20	19
1997	64	32	32
1998	61	26	35
1999	46	29	17
Total	400	202	198

Table IV
Sample Statistics

This table reports size, deal, and target characteristics for the sample. The sample comprises 400 takeovers that were announced over the 1989 to 1999 period. Data are reported for the full sample and for auctions and negotiations. *p*-Value is the *p*-Value from a simple probit regression of the choice of method (i.e., auction vs. negotiation) on the given variable. Panel A reports firm size. *Target Size* and *Bidder Size* are mean (median) equity values in $ billion, measured as (stock price*shares outstanding), estimated 64 days prior to initial announcement. There are only 308 available observations for bidders because 92 bidders were non-U.S. corporations or were private companies. *Relative Size* is target equity value divided by bidder equity value and employs the 308 cases with bidder data. Panel B reports deal characteristics. *Cash* reports the fraction of cases in which the payment was entirely in cash. *Tender* reports the fraction of cases in which the acquisition was by tender offer. *Unsolicited* reports the fraction of cases in which the event was initiated by the bidder or a third party. Panel C reports target characteristics. *Regulated* reports the fraction of cases in which the target was in a regulated industry. *Affiliated* reports the fraction of cases in which the bidder has an ownership affiliation with the target. *Return SD* reports the standard deviation of target stock returns between days −317 and −64 prior to the initial announcement date.

Variable	Full Sample ($N = 400$)		Auction ($N = 202$)		Negotiation ($N = 198$)		*p*-Value
Panel A. Firm Size							
	Mean	Median	Mean	Median	Mean	Median	
Target Size ($ bil)	2.69	0.69	1.68	0.37	3.72	1.07	0.005
Bidder Size ($ bil)	10.58	3.41	6.94	3.03	13.81	4.01	0.004
Relative Size	45%	27%	46%	21%	44%	29%	0.837
Panel B: Deal Characteristics							
% Cash	37%		50%		23%		0.000
% Tender	26%		37%		15%		0.000
% Unsolicited	15%		22%		8%		0.000
Panel C: Target Characteristics							
% Regulated	28%		22%		34%		0.005
% Affiliated	7%		4%		11%		0.011
Return *SD*	0.023		0.024		0.022		0.038

4.4 Table IV: firm, deal, and target characteristics (and a simple probit diagnostic)

Table IV compares auctions and negotiations, and reports a *simple probit* p-value diagnostic for each variable.

Panel A (size). Auction targets are smaller than negotiation targets (target size differs significantly), and bidders in negotiations tend to be larger, while relative size is similar across procedures. **Interpretation:** negotiations tend to occur in "bigger" environments, consistent with lower value dispersion and/or higher information costs of an auction.

Panel B (deal characteristics). Auctions are much more likely to involve cash payment, tender offers, and unsolicited initiation. **Interpretation:** auctions are associated with more "contest-like" deal technologies.

Panel C (target characteristics). Negotiations are more common for regulated targets and for affiliated bidders (pre-existing ownership ties), and auction targets exhibit higher pre-deal return volatility. **Interpretation:** these patterns align with theories in which negotiations mitigate information revelation costs and exploit existing relationships.

5 Main outcome: do auctions create higher target returns? Theory and empirical strategy

5.1 Competing hypotheses

1. **Agency/competition hypothesis:** auctions increase competition and therefore should increase target shareholder gains; negotiations may reflect managerial private benefits.

2. **Information-cost hypothesis:** auctions are costly (information production, potential leakage of proprietary information); hence negotiations can be efficient, and average target gains need not differ.

5.2 Empirical object: abnormal target returns

Let $R_{i,t}$ be the target return and $R_{m,t}$ the market return (CRSP value-weighted index). The paper's main abnormal return measure is **net-of-market**:

$$AR_{i,t} = R_{i,t} - R_{m,t}. \tag{2}$$

For an event window $[\tau_1, \tau_2]$ around the initial announcement date $t = 0$, the cumulative abnormal return is:

$$CAR_i(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}. \tag{3}$$

Main windows:

$$(-1, +1), \quad (-20, +20), \quad (-63, +126).$$

Robustness checks (discussed in text) include market-model returns and buy-and-hold abnormal returns.

6 Model 1: mean comparisons (Tables V and VIII)

6.1 Table V: event-study returns in the short window $(-1, +1)$

Table V reports mean $CAR(-1, +1)$ for auctions vs negotiations. Targets earn about 20–23% around the announcement, but the auction-minus-negotiation difference is *not statistically significant*. **Interpretation:** in the canonical announcement window used in takeover research, auctions do not deliver reliably higher target gains.

Table V

Event Study Analysis: (−1,+1) Window

This table reports event study returns for the full sample of 400 target firms and for the 308 cases in which the bidder is a U.S. publicly traded firm and bidder equity value data are available. The results are net-of-market returns for the (−1,+1) window, where day 0 is the initial announcement date and the market index is the CRSP value-weighted index. The *p*-Values in Panel A and Panel B are for a *t*-test of the null that the mean return equals 0. Panel C presents means tests that test that null that the mean of the two samples are equal (*p*-values are in parentheses).

Sample	Target Return [$N = 400$]	Target Return [$N = 308$]
Panel A: Full Sample—Mean Return (*p*-Value)		
Full Sample	21.6%	20.4%
	(0.000)	(0.000)
Panel B: Analysis by Sales Procedure—Mean Return (*p*-Value)		
Negotiation	20.5%	19.3%
	(0.000)	(0.000)
Auction	22.7%	21.6%
	(0.000)	(0.000)
Panel C: Means Tests—*t*-Statistic (*p*-Value)		
Negotiation V.	1.02	0.90
Auction	(0.308)	(0.368)

Table VIII

Event Study Analysis: Longer Windows

This table reports event study returns for the full sample of 400 target firms and for the 308 cases in which bidder data are available. The results are net-of-market returns for the (−20,+20) and (−63,+126) windows, where day 0 is the initial announcement date and the market index is the CRSP value-weighted index. The *p*-Values in Panel A and Panel B are for a *t*-test of the null that the mean return equals zero. Panel C presents means tests that test that null that the mean of the two samples are equal (*p*-Values are in parentheses).

	Target Return (−20,+20)		Target Return (−63,+126)	
Sample	[$N = 400$]	[$N = 308$]	[$N = 400$]	[$N = 308$]
Panel A: Full Sample—Mean Return (*p*-Value)				
Full Sample	25.4%	23.3%	26.2%	25.1%
	(0.000)	(0.000)	(0.000)	(0.000)
Panel B: Analysis by Sales Procedure—Mean Return (*p*-Value)				
Negotiation	22.6%	21.2%	21.8%	20.9%
	(0.000)	(0.000)	(0.000)	(0.000)
Auction	28.2%	25.7%	30.6%	29.9%
	(0.000)	(0.000)	(0.000)	(0.000)
Panel C: Means Tests—*t*-Statistic (*p*-Value)				
Negotiation V.	2.26	1.68	2.81	2.67
Auction	(0.024)	(0.093)	(0.005)	(0.008)

6.2 Table VIII: longer windows

Table VIII shows that over longer windows, auctions exhibit higher *mean* returns than negotiations, and the difference is statistically significant in simple means tests. **Interpretation:** without controls, auctions look "better" in longer windows, potentially because of differences in deal/target characteristics and timing (runups, leakage, or anticipation).

7 Model 2: single-equation multivariate regressions (Tables VI, IX, XI)

7.1 Baseline OLS model

For a given window, the paper estimates (for deals with bidder equity data, $N = 308$):

$$CAR_i = \alpha + \beta\,\text{Auction}_i + \gamma_1 \ln(\text{RelativeSize}_i) + \gamma_2 \text{Cash}_i + \gamma_3 \text{Tender}_i + \gamma_4 \text{Unsolicited}_i + \varepsilon_i. \quad (4)$$

Here:

- Auction_i is a dummy for an auction sale process,
- $\ln(\text{RelativeSize}_i)$ is log(target equity value / bidder equity value),
- Cash_i is 1 for all-cash deals,
- Tender_i is 1 for tender offers,
- Unsolicited_i is 1 if initiated by bidder/third party.

7.2 Table VI: OLS results for $(-1, +1)$

Table VI estimates (4) in nested specifications.

- The **auction dummy is never statistically significant**, even after adding controls.
- ln(RelativeSize) is **negative and significant**: smaller targets relative to bidders have larger announcement gains.

- **Unsolicited** deals have higher target returns (positive and significant).
- Cash and tender offer indicators matter in bivariate settings but lose significance in the richest specification.

Interpretation: after controlling for standard determinants of takeover announcement returns, auctions do not generate systematically higher target gains.

Table VI
Multiple Regression Analysis: (−1,+1) Window

This table reports regression analysis of target returns on variables for the sales procedure, relative deal size, the payment method, and the acquisition form. *Auction* is a dummy variable equal to 1 when the sales procedure is an auction. *Relative Size* is the natural log of the equity value of the target divided by the equity value of the bidder 64 days prior to the initial announcement date. *Cash* is a dummy variable equal to 1 for acquisitions in which the payment is entirely in cash. *Tender* is a dummy variable equal to 1 when the form of the acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals that were initiated by the bidder or a third party. The regressions use the 308 acquisitions with available bidder equity data. The dependent variable in each regression is net-of-market returns for the (−1,+1) window, where day 0 is the initial announcement date and the market index is the CRSP value-weighted index. The *p*-values of the regression coefficients are reported in parentheses.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	0.193	0.133	0.179	0.184	0.185	0.119
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Auction	0.023	0.008	0.003	0.008	0.004	−0.017
	(0.360)	(0.739)	(0.914)	(0.746)	(0.862)	(0.487)
Relative size	–	−0.050	–	–	–	−0.047
		(0.000)				(0.000)
Cash	–	–	0.089	–	–	0.035
			(0.002)			(0.430)
Tender	–	–	–	0.085	–	0.014
				(0.009)		(0.778)
Unsolicited	–	–	–	–	0.105	0.091
					(0.003)	(0.009)
Adjusted R^2	0.000	0.082	0.026	0.018	0.026	0.108
Model *p*-Value	0.360	0.000	0.006	0.022	0.007	0.000

Table IX
Multiple Regression Analysis: Longer Windows

This table reports regression analysis of target returns on variables for the sales procedure, relative deal size, the payment method, and the acquisition form. *Auction* is a dummy variable equal to 1 when the sales procedure is an auction. *Relative size* is the natural log of the equity value of the target divided by the equity value of the bidder 64 days prior to the initial announcement date. *Cash* is a dummy variable equal to 1 for acquisitions in which the payment is entirely in cash. *Tender* is a dummy variable equal to 1 when the form of the acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals that were initiated by the bidder or a third party. The regressions use the 308 acquisitions with available bidder equity data. The dependent variable in each regression is net-of-market returns for the (−20,+20) and (−63,+126) windows, where day 0 is the initial announcement date and the market index is the CRSP value-weighted index. The *p*-values of the regression coefficients are reported in parentheses.

Variable	(−20,+20)	(−63,+126)
Intercept	0.117	0.111
	(0.000)	(0.000)
Auction	−0.004	0.030
	(0.876)	(0.368)
Relative size	−0.061	−0.061
	(0.000)	(0.000)
Cash	0.022	−0.013
	(0.638)	(0.825)
Tender	0.037	0.050
	(0.479)	(0.447)
Unsolicited	0.115	0.209
	(0.002)	(0.000)
Adjusted R^2	0.160	0.152
Model *p*-Value	0.000	0.000

7.3 Table IX: OLS results for longer windows

Table IX repeats the regression for $(-20, +20)$ and $(-63, +126)$ windows. Again, the **auction coefficient is statistically insignificant**. **Interpretation:** the mean differences in Table VIII do not survive conditioning on observable deal characteristics.

7.4 Table XI: OLS results for the stock-payment subsample

For the subsample of deals using some or all stock as payment ($N = 228$), the paper estimates an OLS model analogous to (4) but replaces Cash with AllStock:

$$CAR_i = \alpha + \beta\,\text{Auction}_i + \gamma_1 \ln(\text{RelativeSize}_i) + \gamma_2 \text{AllStock}_i + \gamma_3 \text{Tender}_i + \gamma_4 \text{Unsolicited}_i + \varepsilon_i. \quad (5)$$

Table XI shows:

- auction remains positive but **insignificant** across all windows,
- all-stock payment is **insignificant**,
- relative size is negative and significant; unsolicited is positive for longer windows.

Interpretation: even within the stock-payment setting (where legal constraints differ), auctions do not deliver reliably higher target gains.

Table VII

Two-Stage Regression Analysis: (−1,+1) Window

This table reports simultaneous equations analysis for target returns and for the choice of sales procedure. Returns are measured using net-of-market returns for the (−1,+1) window, where day 0 is the announcement date and the market index is the CRSP value-weighted index. *Procedure** is the predicted value of the choice of sales procedure. *Returns** is the predicted value of returns. *Target Size* is the equity value of the target and *Relative Size* is the natural log of the equity value of the target divided by the equity value of the bidder, both 64 days prior to the announcement date. *Cash* is a dummy variable equal to 1 when the payment is entirely in cash. *Tender* is a dummy variable equal to 1 when acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals that were initiated by the bidder or a third party. *Regulated* is a dummy variable equal to 1 for targets in regulated industries. *Affiliated* is a dummy variable equal to 1 for targets having an ownership affiliation with the bidder. *Return SD* is the standard deviation of target stock returns between days −317 and −64 prior to the initial announcement date. The analysis uses the 308 takeovers with available bidder equity data (*p*-values are in parentheses).

	Target Returns		Sales Procedure	
Variable	1st Stage	2nd Stage	1st Stage	2nd Stage
Intercept	0.090	0.129	2.386	2.439
	(0.512)	(0.002)	(0.009)	(0.012)
Procedure*		−0.016		
		(0.732)		
Returns*				−0.583
				(0.726)
Relative Size	−0.040	−0.039	0.023	
	(0.000)	(0.000)	(0.727)	
Target Size	0.003		−0.184	−0.182
	(0.733)		(0.001)	(0.001)
Cash	0.039	0.047	0.508	0.530
	(0.380)	(0.370)	(0.077)	(0.082)
Tender	−0.013	−0.013	−0.024	−0.032
	(0.794)	(0.788)	(0.943)	(0.925)
Unsolicited	0.084	0.095	0.676	0.725
	(0.013)	(0.048)	(0.003)	(0.006)
Regulated	−0.073	−0.074	−0.084	−0.127
	(0.009)	(0.009)	(0.635)	(0.670)
Affiliated	−0.042	−0.049	−0.464	−0.488
	(0.417)	(0.390)	(0.185)	(0.170)
Return *SD*	0.990	0.918	−4.454	−3.876
	(0.452)	(0.460)	(0.602)	(0.661)
Adjusted R^2	0.126	0.126	0.112	0.112
Model *p*-Value	0.000	0.000	0.000	0.000

Table XI

Multiple Regression Analysis: Some or All Stock as Payment Method

This table reports regression analysis of target returns for the subsample of deals using some or all stock as the method of payment. *Auction* is a dummy variable equal to 1 when the sales procedure is an auction. *Relative Size* is the natural log of the equity value of the target divided by the equity value of the bidder 64 days prior to the initial announcement date. *All Stock* is a dummy variable equal to 1 when the payment is 100% in stock. *Tender* is a dummy variable equal to 1 when the form of the acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals that were initiated by the bidder or a third party. The regressions use the 228 acquisitions with available bidder equity data. The dependent variable in each regression is net-of-market returns for the each window, where day 0 is the initial announcement date and the market index is the CRSP value-weighted index. The *p*-Values of the regression coefficients are reported in parentheses.

Variable	(−1,+1)	(−20,+20)	(−63,+126)
Intercept	0.085	0.078	0.071
	(0.013)	(0.029)	(0.145)
Auction	0.036	0.037	0.061
	(0.194)	(0.200)	(0.130)
Relative size	−0.056	−0.072	−0.076
	(0.000)	(0.000)	(0.000)
All stock	0.010	0.017	0.016
	(0.738)	(0.593)	(0.709)
Tender	−0.059	−0.120	−0.422
	(0.762)	(0.557)	(0.132)
Unsolicited	0.030	0.092	0.200
	(0.504)	(0.047)	(0.002)
Adjusted R^2	0.096	0.160	0.152
Model *p*-Value	0.000	0.000	0.000

8 Model 3: endogeneity of procedure choice (Tables VII, X, XII)

8.1 Why endogeneity matters

The procedure choice is a *decision variable*: targets choose auction vs negotiation partly based on expected surplus. If so, Auction_i is endogenous in (4) and OLS may be inconsistent.

8.2 Simultaneous system

The paper models:

$$\textbf{(Return equation)} \qquad CAR_i = \alpha + \beta\,\text{Auction}_i + X_i'\gamma + \varepsilon_i, \tag{6}$$

$$\textbf{(Procedure equation)} \qquad \text{Auction}_i = \mathbf{1}\{A_i^* > 0\}, \qquad A_i^* = \delta + Z_i'\theta + \lambda\, CAR_i + u_i. \tag{7}$$

Here X_i includes standard deal controls (cash, tender, unsolicited, regulated, affiliated, return volatility, etc.), and the procedure equation is a **probit** model.

8.3 Identification via exclusion restrictions

To identify the system, the paper excludes one exogenous variable from each structural equation:

- **Exclude target size from the return equation:** target size helps explain procedure choice (auction vs negotiation) but is not robustly related to announcement returns.
- **Exclude relative size from the procedure equation:** relative size predicts target returns but (per Table IV) is not related to procedure choice.

Table X

Two-Stage Regression Analysis: Longer Windows

This table reports simultaneous equations analysis for target returns and for the choice of sales procedure. Returns are measured using net-of-market returns for the (−20,+20) and (−63,+126) windows, where day 0 is the announcement date and the market index is the CRSP value-weighted index. *Procedure** is the predicted value of the choice of sales procedure. *Returns** is the predicted value of returns. *Target Size* is the equity value of the target and *Relative Size* is the natural log of the equity value of the target divided by the equity value of the bidder, both 64 days prior to the initial announcement date. *Cash* is a dummy variable equal to 1 when the payment is entirely in cash. *Tender* is a dummy variable equal to 1 when the form of the acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals that were initiated by the bidder or a third party. *Regulated* is a dummy variable equal to 1 for targets in regulated industries. *Affiliated* is a dummy variable equal to 1 for targets having an ownership affiliation with the bidder. *Return SD* is the standard deviation of target stock returns between days −317 and −64 prior to the initial announcement date. The analysis uses the 308 takeovers with available bidder equity data (p-values are in parentheses).

	Target Returns (−20,+20)		Target Returns (−63,+126)	
Variable	1st Stage	2nd Stage	1st Stage	2nd Stage
Intercept	0.044	0.094	0.047	0.054
	(0.764)	(0.032)	(0.802)	(0.326)
Procedure*		−0.021		−0.003
		(0.675)		(0.959)
Relative size	−0.052	−0.052	−0.051	−0.051
	(0.000)	(0.000)	(0.000)	(0.000)
Target size	0.004		0.001	
	(0.675)		(0.959)	
Cash	0.029	0.039	−0.001	0.001
	(0.533)	(0.474)	(0.989)	(0.990)
Tender	0.015	0.014	0.039	0.039
	(0.778)	(0.786)	(0.559)	(0.560)
Unsolicited	0.113	0.127	0.218	0.220
	(0.002)	(0.013)	(0.000)	(0.001)
Regulated	−0.057	−0.059	−0.013	−0.013
	(0.050)	(0.049)	(0.730)	(0.731)
Affiliated	−0.072	−0.082	−0.052	−0.053
	(0.182)	(0.176)	(0.451)	(0.488)
Return *SD*	2.580	2.487	3.986	3.973
	(0.064)	(0.059)	(0.024)	(0.018)
Adjusted R^2	0.183	0.183	0.166	0.166
Model p-Value	0.000	0.000	0.000	0.000

Table XII

Two-Stage Regression Analysis: Some or All Stock as Payment Method

This table reports simultaneous equations analysis for target returns for the subsample of deals using some or all stock as payment. Returns are measured using net-of-market returns, where day 0 is the announcement date and the market index is the CRSP value-weighted index. *Procedure** is the predicted value of the choice of sales procedure. *Target Size* is the equity value of the target and *Relative Size* is the natural log of the equity value of the target divided by the equity value of the bidder, both 64 days prior to the initial announcement date. *All Stock* is a dummy variable equal to 1 when the payment is 100% in stock. *Tender* is a dummy variable equal to 1 when the acquisition is a tender offer. *Unsolicited* is a dummy variable equal to 1 for deals initiated by the bidder or a third party. *Regulated* is a dummy variable equal to 1 for targets in regulated industries. *Affiliated* is a dummy variable equal to 1 for targets having an ownership affiliation with the bidder. *Return SD* is the standard deviation of target stock returns between days −317 and −64 prior to the announcement date. The analysis uses the 228 takeovers with available bidder equity data (p-values are in parentheses).

	Returns (−1,+1)		Returns (−20,+20)		Returns (−63,+126)	
Variable	1st Stage	2nd Stage	1st Stage	2nd Stage	1st Stage	2nd Stage
Intercept	0.199	0.105	0.141	0.079	0.124	0.017
	(0.178)	(0.047)	(0.360)	(0.127)	0.563	(0.811)
Procedure*		0.066		0.043		0.075
		(0.479)		(0.642)		(0.568)
Relative size	−0.046	−0.045	−0.063	−0.062	−0.067	−0.066
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)
Target size	−0.007		−0.005		−0.008	
	(0.444)		(0.633)		(0.553)	
All stock	0.020	0.051	0.025	0.045	0.017	0.052
	(0.517)	(0.379)	(0.433)	(0.432)	(0.692)	(0.515)
Unsolicited	0.044	−0.033	0.113	0.063	0.241	0.154
	(0.302)	(0.785)	(0.011)	(0.601)	(0.000)	(0.367)
Regulated	−0.067	−0.064	−0.054	−0.052	0.007	0.011
	(0.016)	(0.038)	(0.061)	(0.087)	(0.854)	(0.796)
Affiliated	−0.049	−0.012	−0.114	−0.090	−0.096	−0.054
	(0.384)	(0.884)	(0.054)	(0.273)	(0.247)	(0.643)
Return *SD*	1.434	1.312	2.097	2.018	3.999	3.862
	(0.310)	(0.411)	(0.154)	(0.202)	(0.052)	(0.084)
Adjusted R^2	0.129	0.129	0.193	0.193	0.155	0.155
Model p-Value	0.000	0.000	0.000	0.000	0.000	0.000

8.4 Estimation: two-stage probit least squares (2SPLS)

The implementation is:

1. First stage: regress CAR_i on the full set of exogenous variables (including excluded instruments) to get $\widehat{CAR}_i$; estimate the probit for procedure on the full exogenous set to get $\widehat{\text{Auction}}_i$ ("Procedure*").
2. Second stage: use $\widehat{\text{Auction}}_i$ in the return equation and $\widehat{CAR}_i$ in the procedure equation, with corrected standard errors for the probit-based fitted regressor.

8.5 Table VII: two-stage results for $(-1, +1)$

Table VII reports both first- and second-stage regressions.

- In the **return equation**, the fitted procedure term (Procedure*) is **insignificant**.
- In the **procedure equation**, target size is **negative and significant** (larger targets are less likely to be sold via auction), while relative size is insignificant for procedure choice (consistent with the exclusion idea).

Interpretation: after correcting for endogeneity, auctions and negotiations still generate similar target gains in the short window.

8.6 Table X: two-stage results for longer windows

Table X repeats the two-stage analysis for $(-20, +20)$ and $(-63, +126)$. Procedure* remains **statistically insignificant**. **Interpretation:** the "auction premium" suggested by unconditional longer-window means is not robust to endogeneity correction.

8.7 Table XII: two-stage results for stock-payment subsample

Table XII applies two-stage analysis within the stock-payment subsample. Again, Procedure* is **insignificant** across all event windows. **Interpretation:** the core conclusion is stable across payment-method subsamples.

9 Overall findings (tie all tables/figures together)

9.1 What the process data say about competition

1. The private phase is deep: many contacts, multiple confidentiality agreements, and private bids (Tables I–II).
2. Public bids are rare: most deals have a single public bidder even when the private process is competitive (Table II).
3. Therefore, measuring competition by public bids severely understates true competition (Figure 1 + Tables I–II).

9.2 What the outcome analysis says about auctions vs negotiations

Across short and long windows, and across OLS and endogeneity-corrected specifications:

Target returns are broadly similar in auctions and negotiations.

This evidence supports a view in which procedure choice trades off competition against transaction/information costs, rather than reflecting systematic managerial diversion of value through negotiation.

10 Short summary

1. **Measurement lesson:** public-bid counts are a noisy proxy for competition; private process data matter.
2. **Classification:** auctions are defined by contacting multiple potential buyers (and multiple confidentiality agreements), not by public bidding wars.
3. **Event-study result:** in the canonical $(-1, +1)$ window, auctions do not have higher target gains (Table V).
4. **Regression result:** auction dummy is insignificant with standard controls (Table VI) and for longer windows (Table IX).
5. **Endogeneity correction:** two-stage probit least squares still finds no auction premium (Tables VII, X, XII).

11 Conclusion

11.1 Summary

Boone and Mulherin (2007) conclude that **how firms are sold is largely determined in a private takeover process that is typically invisible in public bid data**, and that once this private process is measured directly, two widely held inferences in the M&A literature must be re-evaluated.

(i) **Public takeover activity severely understates true takeover competition.** Counting *public* bidders gives a misleadingly low view of competition because most rivalry occurs *privately*: targets (and their advisors) contact many potential acquirers, sign multiple confidentiality/standstill agreements, and often solicit private written bids, even when only one bidder ever makes a public offer.

(ii) **A substantial share of large corporate sales are auctions, even if public bidding wars are rare.** Using the private-process classification, about **half** of targets in the sample are sold via auctions (defined by contacting multiple potential buyers), which implies much more competition than suggested by the frequency of multi-bidder public contests.

(iii) **Auctions do not robustly generate higher target shareholder wealth effects than negotiations.** In the canonical short announcement window, mean target abnormal returns are similar for auctions and negotiations, and the difference is statistically insignificant. While longer-window means are higher for auctions, this "auction premium" disappears in multivariate regressions with standard deal controls and in two-stage specifications that account for the endogeneity of the procedure choice.

(iv) **Procedure choice is systematic and consistent with an efficiency trade-off, not primarily agency avoidance of competition.** Auctions and negotiations are chosen in predictable ways related to target size and other characteristics. The evidence fits models in which auction versus negotiation reflects a trade-off between (a) the benefits of competition and (b) information/transaction costs, and is inconsistent with the view that negotiations are systematically used to reduce competition and harm shareholders.

11.2 Key facts and magnitudes

These numbers are the cleanest "memory anchors" from the paper:

- **Sample size:** 400 major U.S. takeovers (1989–1999).
- **How much competition is visible publicly?** Average *public* bidders is only **1.13**, and only **51 out of 400** deals have more than one public bidder.
- **How much competition occurs privately?** In the full sample, targets contact on average **9.49** potential buyers, obtain **3.75** confidentiality agreements, and receive **1.29** private written bids (Table II, Panel A).

- **Auction vs negotiation intensity (private stage):**
  - Auctions ($N = 202$): Contact $\approx$ **20.67**, Confidential $\approx$ **6.83**, Private bids $\approx$ **1.57**, Public bidders $\approx$ **1.24**.
  - Negotiations ($N = 198$): Contact $\approx$ **1.02**, Confidential = **1**, Private bids = **1**, Public bidders $\approx$ **1.02**.
- **Target announcement returns:** $CAR(-1, +1)$ is large in both procedures: full-sample means are about **20.5%** (negotiation) vs **22.7%** (auction), and the difference is not statistically significant (Table V).
- **Endogeneity-corrected evidence:** In the two-stage system, the fitted procedure term (Procedure*) is economically small and statistically insignificant in the return equation (e.g., -0.016 with $p = 0.732$ in the $(-1, +1)$ window; Table VII), and the same qualitative conclusion holds for longer windows (Tables X and XII).

11.3 Why auctions and negotiations deliver similar target returns

The central intuition is a **selection-and-cost trade-off**.

1) Auctions raise competition but also raise information and transaction costs. Running an auction requires contacting many potential buyers and sharing sensitive information with multiple parties. This can increase the expected sale price through competition, but it can also: (i) increase the cost of information production (data room, diligence, bargaining), (ii) increase leakage risk (loss of proprietary information), and (iii) create execution risk (delay, complexity, and strategic behavior by bidders).

2) Negotiations can be efficient when the marginal value of extra competition is low or the marginal cost is high. If the target is large, regulated, or has a natural affiliated buyer, then the incremental benefit from broad competition may be limited, while the costs of exposing information broadly may be high. In such cases, negotiating with one bidder can maximize expected value for shareholders.

3) A negotiation is not "competition-free" because the target can credibly threaten to open an auction. Because the private process is well developed, the target and its advisor can solicit competing bids if negotiations stall. This credible outside option disciplines the negotiating bidder and helps explain why negotiated sales can generate returns comparable to auctions.

11.4 Methodological intuition: what this paper changes in empirical M&A work

The paper's core methodological contribution is that **competition must be measured in the private stage, not inferred from public bidder counts**. Public bid data capture only late-stage outcomes, not the earlier search, screening, and bidding rounds. Therefore:

- "one public bidder" does *not* imply "no competition," and

- empirical conclusions about the effects of "auctions" versus "negotiations" can be biased if the procedure is misclassified or treated as exogenous.

11.5 Takeaway

Public bids are only the tip of the iceberg: most takeover competition occurs privately, about half of major firms are sold via auctions in that private stage, and once procedure choice and endogeneity are accounted for, auctions do not reliably deliver higher target shareholder returns than negotiations—consistent with an efficient trade-off between competition and information/transaction costs rather than systematic agency avoidance of auctions.